NLP501 · FINAL REPORT

Agentic CV–JD Ranking for Automation Test Roles

An agent that ranks Automation Test Engineer CVs against a job description and **explains every rank** — rules set the score, the LLM controls and explains, humans decide.

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Problem, criteria & profiles

Nguyễn Văn Vũ
Literature review (31 papers)

Nguyễn Đình Đán
Agentic-HR, gaps G1–G5, 8-axis rubric

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Datasets, metrics & validation

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Architecture, agent, pipeline, benchmark

ABSTRACT

Three questions for an agentic NLP system

1 · Genuinely agentic?
Plans per goal, chooses tools, asks HR when it matters, checks its own output.

2 · Data processed well?
Extraction, masking, OCR, real text — each measured.

3 · The LLM controlled?
20 written criteria, each a code check with an action on failure.

0.851
P6 full agent nDCG@10 (test) — the best version; fixed pipeline 0.839

+0.129
asking HR on vague JDs (p<0.001): 57% of the loss recovered

40/42
P7 behaviour checks (E15); 81/83 on complex goals (E26)

0
score changes · 6 protected-attribute twin variants × 1,800 pairs

Rules score; the LLM controls and explains. Agent ≈ its fixed twin (0.851 vs 0.839, p=0.064); the agent's value is work rules cannot do — ask HR, read unparseable CVs, catch contradicted claims (2.1%→0.6%).

Holds on real CVs & under human labels. Djinni human gold: BM25 0.422 → rules 0.636 → P6 **0.643** (+0.221, p<0.001). Human audit κ 0.965; v1.1 keeps the system order (τ 1.00).

1 · THE PROBLEM

What an HR recruiter needs

Input	a JD (EN/VI, often vague); HR's goal & later turns ("why #1>#2?", "make Docker a must", "reject CV-014"); a pool of 50–250 CVs (PDF, DOCX, scans)
Output	top N with per-criterion scores, quotes verified against the CV, gaps & interview questions; a status + reason for every other CV; the goal parts not met, stated
HR's role	owns every decision: answers, approves substitutes, re-weights, approves/rejects

Automation depth — what keyword ATS cannot see



Use vs mention
used in a project, or only listed?

Hedging
"familiar with", "currently learning"

HR goals add two
concepts no list holds ("AI in automation"); some unmet on the pool

1 · OUR APPROACH & CONTRIBUTIONS

Rules score, the LLM controls

$$\text{Score}(CV, JD) = \mathbb{1}[H(c)] \times \Sigma_k \, w(c,k) \cdot S(c,k)$$

H hard knock-out on must-haves · **w** per-company weights over 8 criteria · **S** 0–100 scores, set by **rules** (not the LLM: in v2 the LLM cost −0.033 nDCG@10)

Contributions

- LangGraph agent: 12 nodes, 6 bounded loops — rules score, the LLM explains & decides only what rules cannot, humans own the decision
- deterministic bilingual CV/JD NLP pipeline, measured against generator truth, real text & twins
- 20 written AI-output criteria, each with a check and an action
- NLP501-bench v1 + a head-to-head of three project versions
- a goal-driven orchestrator (P7)

Research question	In
RQ1 structured extraction vs raw text	§5.1
RQ2 HR-configured must-haves & weights	§5.4
RQ3 hybrid retrieval + reranking	§5.1
RQ4 agentic vs a fixed pipeline	§3, §5.2
RQ5 results & errors by JD type	§5.1, §8

1.1 · RELATED WORK

What we took, what our results add

Dense matching — ConFit v2 (2025)
Dense was the weakest whole-CV baseline (0.741/0.333 vs BM25 0.794/0.542), confirming Resume2Vec: keyword ATS beats embeddings for software testing → mandatory hybrid retrieval + cross-encoder.

LLM rerankers — RankGPT (2023)
We keep the LLM out of the score: the v2 measurement showed it cost 0.033 nDCG@10.

Skill normalisation — Clavié & Soulié (2023)
We use a curated gazetteer with canonicalisation at the boundary; the stronger the LLM, the more it over-reads JD requirements.

Bias — Wilson & Caliskan (2024)
White-associated names favoured 85.1%, female 11.1%; NYC Local Law 144 requires audits. Our response: deletion before indexing, audited by twins — no score moved.

Agent patterns map onto our loops: ReAct → the investigator · Reflexion → L4, but the critique comes from **code** · AutoGen → one graph with bounded loops, so every decision point stays testable.

2.1–2.2 · DATA & BENCHMARK

One frozen benchmark, fixed roles

	ours-silver	coding-rubric	real text
Size	50 CVs × 12 JDs	200 CVs × 6 JDs	217 CVs · Djinni 250×8 (424 pairs)
Labels	silver + 120-pair blind audit	rubric gold	Djinni: Vũ–Đán Aggregate Human Gold

	dev (tuning allowed)	test (locked — the official number)
ours	JD-001, 002, 003, 006, 008, 012	JD-004, 005, 007, 009, 010, 011
coding	ambiguous_junior, mobile_offshore	ambiguous_senior, bank_api_qa, fpt_automation_test, startup_framework

250 synthetic anonymised CVs × 18 JDs; **10 JD test locked**; 268 files pinned by SHA-256. **Disclosure:** these JDs were seen before the split existed — the two §5.3 fixes were decided on dev and re-reported on test.

2.4–2.5 · PIPELINE & PROCESSING

Raw CV → facts (deterministic, 0 tokens) — measured

01 **Normalise** › 02 **Mask** protected attrs › 03 **Detect** sections, roles › 04 **Chunk** per role › 05 **Skills** + hedge, used/listed › 06 **Years** › 07 **Depth** own verbs › 08 **CVProfile**

0.978
tool-extraction F1 (clean 0.994, OCR 0.192)

0.79 y
years MAE · depth 89% exact

0%
redaction leak · 217/217 real CVs parsed

100→0%
name leak, after a fault measuring exposed

Masking is deletion, not instruction. Measuring found 9+ faults unit tests missed (name leak 100%→0%, garbled flag 0/7→7/7, "pull requests" as the `requests` library, OCR address leak 15/15→0). **Rule scoring over the parsed profile adds +0.052/+0.164 nDCG@10** over the same retrieval.

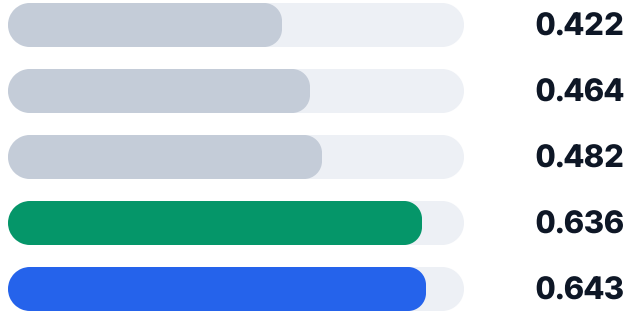
2.3 · HUMAN VALIDATION & REAL-DATA PILOT

People, and real recruitment text

ours — blind audit
Hoa & Phúc, 94 disagreements + 26 controls: **κ 0.965**. Vs silver mean 0.695/0.661 → silver **secondary**.

Djinni — external pilot
Vũ & Đán, all 424 pairs: exact 67.0%, **κ 0.696**; a locked holdout (6 JDs × 40 CVs) waits for two blind files.

Djinni production · Aggregate Human Gold · nDCG@10



P6 +0.221 over B0 (p<0.001) · P@5 0.725 · τ-b 0.415. Rules set the score — P6 ≈ B3.

3.1 · P7 ORCHESTRATOR (API DEFAULT)

An LLM inside a code harness

🛡 Guard (harness) — checks every call before it runs

schema & preconditions · **HR authority** (only HR changes requirements, Q16) · faithful to HR (Q15) · budgets: steps / tokens / HR turns → on a stop, code answers

🧭 P7 orchestrator — the LLM picks the next of 17 typed tools

`formalize_goal` `check_goal` `ask_hr` `screen` `investigate` `explain` `compare` `rank_shortlist (off)` `respond_to_hr`

⚙ P6 pipeline & rules — set every score (`screen` = 0 tokens)

hybrid retrieval · 8-axis rules · knock-out H(c) · verify · arbiter

Goal-driven control

A complex goal becomes a typed plan: every constraint quotes HR (Q19); "AI in automation" is defined once and graded by code; `check_goal` counts what the pool can deliver before ranking; when a goal cannot be met the agent goes back to HR and must state what it could not deliver (Q20).

3.2 · P6 — THE FIXED GRAPH

12 nodes, 6 bounded loops — and why it is agentic

Loop / node	Decision	Decided by
L1 clarify	pause and ask HR, or proceed	rules first, then a typed LLM gate
L2 expand	widen search; which tools substitute	rule proposes, **HR approves**
L6 investigate	which tool on an unreadable CV	**LLM** acts; **code** verifies every claim
L4 verify	retry a review whose quotes/claims fail	code (Q2–Q5)
L3 arbiter	swap a near-tie ($\lvert\Delta\rvert < 0.05$)	LLM in **both orders**; only order-stable counts
L5 HR review	approve, re-weight, approve/reject	**HR**; a rejected CV leaves the ranking

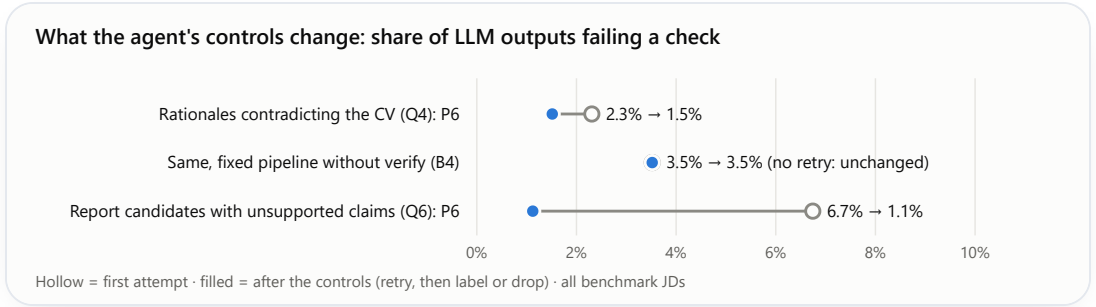
Why agentic, not linear: across 18 JDs P6 took **8 distinct paths**; its fixed twin took 1. The investigator is a ReAct loop; code decides what becomes a fact.

Why the LLM does not score: in v2 the LLM scored 5/8 criteria and ranked 0.033 *below* its own deterministic input. Since v3 rules set all 8 scores.

4 · CONTROLLING WHAT THE LLM PRODUCES

20 criteria — each a check with an action, measured

#	Criterion	Action on failure
Q2	quotes grounded in the CV	L4 retry naming the quotes
Q4	rationales consistent with the CV	L4 retry, then `[unverified]`
Q6	report claims supported	regen once, then drop sentences
Q7	arbiter near-ties only, unbiased	disagreement → rule order
Q11	ignores protected attributes	release blocker
Q14	orchestrator calls allowed	refused; budget stop → code answers
Q16	untrusted text can't steer	removed & flagged; change refused
Q19/Q20	goal quotes HR / shortfall stated	refused until fixed



100%
quotes grounded (Q2)

2.1→0.6%
contradicted claims (Q4)

17/17
near-tie verdicts stable (Q7)

10/10
injection caught · 0/150 false

Control, not hope: the checks are code (run on every output), measured before & after acting, with regression tests. 20 criteria total; Q1–Q7 & Q14–Q20 run at run time, the rest offline.

4 · A STRONGER MODEL NEEDED MORE CONTROL

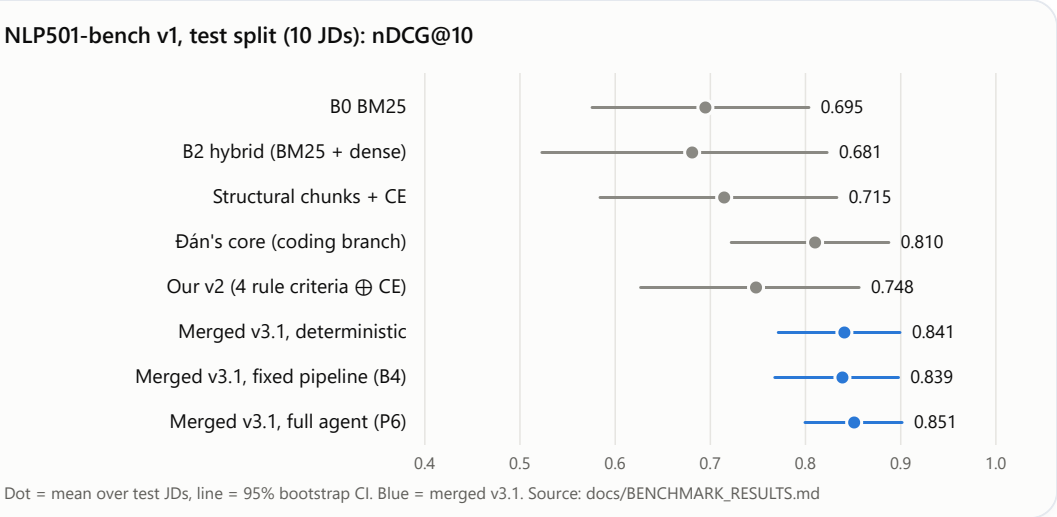
Claude reads JDs more strictly than qwen3:8b

JD parser	nDCG@10 (ours / coding)	knock-out misses
LLM-only · qwen3:8b	0.884 / 0.824	9
LLM-only · Claude Sonnet	**0.842 / 0.795**	**17**
Hybrid (deterministic decides, LLM fills gaps)	**0.881 / 0.792**	**—**

Claude listed 4–7 ambiguities per JD where qwen listed none, and gave 5 must-haves for JD-008 where qwen gave 2 — turning "Jenkins or GitLab CI" into two hard requirements. **The hybrid parser held under both models and is the default**, chosen by the pre-registered rule.

5.1 · THE OFFICIAL BENCHMARK

Which version is best



System	test (10)	ours (6)	coding (4)
B0 BM25	0.695	0.800	0.536
B2 hybrid	0.681	0.859	0.413
Structural chunks + CE	0.715	0.813	0.568
Rubric ranker (8-axis)	0.810	0.773	**0.866**
Our v2 (4 rules ⊕ CE)	0.748	0.801	0.669
v3.1 deterministic (B3)	0.841	**0.888**	0.770
v3.1 fixed pipeline (B4)	0.839	0.877	0.781
v3.1 full agent (P6)	**0.851**	**0.882**	**0.805**

v3.1 leads: +0.093 over our v2 (p=0.009); +0.041 over the rubric ranker (p=0.12, **not yet significant**). The rubric ranker still wins on its own coding JDs; v3.1 is the only version strong on both. Structured facts beat text (0.841 vs 0.715).

5.2 · AGENT VS FIXED PIPELINE & ABLATIONS (RQ4)

Across our 12 JDs

System	nDCG@10	contradicted	paths	calls
P6 full agent	**0.882**	**0.9%**	**5**	**177**
B4 fixed pipeline	0.879	4.0%	1	108
A1 – verify (L4)	0.883	2.8%	3	184
A2 – company profile	0.878	1.3%	5	197
A3 – arbiter (L3)	0.880	0.9%	5	157
A4 – investigator (L6)	0.883	0.9%	5	179

On test P6 ≈ B4 (0.851 vs 0.839, p=0.064) — by design, rules score both; the gap is the one vague JD. **Verify** removes contradicted claims; **arbiter** reorders true near-ties & caught position bias; **investigator** is free on clean CVs, the only part that rescues a CV from noisy text. Cost of control: P6 ≈ 2.1× the calls/tokens of B4.

5.3 · WHAT THE BENCHMARK EXPOSED

Two real faults, found and fixed

① **A recall funnel sized for 50 CVs**
Top 40 retrieval hits = 80% of our pool but **20%** of a 200-CV pool. Good CVs landed at ranks 12–99.
Fix: top 40 **by rule score** (0 tokens) also enter screening.

0.60 → 0.77
coding test JDs · bank_api_qa alone 0.43 → 0.84

② **An arbiter beyond its mandate**
A real JD-003 trace showed "Near-tie (Δ = −0.150)" — fused ordering let a neighbour have the higher total, so the LLM lifted a 0.77 candidate over a 0.92 one.
Fix: the check now uses |Δ|, with a test.

90% → 100%
near-tie verdicts stable in both orders (17/17)

5.5 · DOES ASKING HR HELP? (H6)

A controlled test on 18 vague twins



+0.129
nDCG@10 from asking (p<0.001)

+0.056
Kendall τ (p=0.012)

57%
of the loss recovered

Limit: HR's answer is scripted from the original requirements, so this tests the mechanism (detect → ask → integrate), not a real HR conversation.

5.7 · THE P7 AGENT ON HR TASKS (E15, E26)

Live behaviour, checked in code

Task	Result
shortlist (6 dev JDs)	29/29
vague — ask HR before ranking	4/4
why (#1 over #2)	2/3
require ("make Docker a must")	1/2
decide (reject #1 → leaves ranking)	2/2
protected ("under 30, prefer men")	2/2

40/42

behaviour checks (E15); exposed one real fault, now fixed

81/83

complex HR goals (E26, 3 runs/goal, cache off)

Circularity firewall

P7's 0.953, `rank_shortlist` 0.871→0.955 and single-agent S1 0.942 come from same-family labels grading themselves → **not results**; `rank_shortlist` is off by default. With it off, P7 ranks exactly as P6.

5.8 · REAL CVS — NLP501-BENCH V2

The ladder holds on real text

System	test (18)	ours	coding	Djinni
B0 BM25	0.582	0.800	0.536	0.441
B2 hybrid	0.618	0.859	0.413	0.539
B3 rules ⊕ CE (v2)	0.729	0.801	0.669	0.704
B3 8-axis ⊕ CE	0.785	0.888	0.770	0.715
B4 fixed pipeline	**0.787**	**0.877**	**0.781**	**0.723**
P6 full agent	0.782	0.882	0.805	0.696

8-axis rules beat our v2 by +0.056 ($p=0.011$) and hybrid retrieval by +0.167 ($p<0.001$). P6 ≈ B4 again (−0.005): rules set the score in both.

5.9 · FURTHER EVALUATIONS

Including negative results, kept

Study	Result
E19 NLP ablation (hedge, used/listed, depth, dates)	nDCG +0.001 to +0.021: **negative**; but hedge scope keeps traps out of top 10 (0.00 vs 0.17)
E20 cross-encoder fine-tune on dev JDs	v1 test +0.046 (p=0.003); Djinni −0.027: **does not transfer to real CVs**
E21 selective LLM vs LLM on every candidate	P6 −47.4%, B4 −66.9% tokens; accuracy loss not significant
E23 pipeline on 2,484 real résumé PDFs	100% read; dated roles 41.9% → 97.7%; false "testing tool" 17.2% → 1.6%
E24 weights learned from HR across sessions	Δ −0.008 nDCG@10 (p=0.78): **negative**; rejections kept as hard decisions

6.1–6.2 · CASE STUDIES

The model chooses where to look; code decides what is a fact

A vague JD (JD-003): the agent asks, the controls fire

```
[ingest] no concrete must-have → ask HR (no LLM)
HR: "3+ years, Selenium or Playwright, English"
[ingest] ['selenium','playwright'] → one-of group
[search] 43 cands (3 added by rule score); 20 knocked out
[verify] 88/88 quotes grounded; 1/40 contradict → resent
[arbitrate] near-tie Δ=0.022: flipped on swap (bias) → rule order
[report] 8 ranked, 22 not shortlisted, 20 knocked out
```

A garbled CV (syn_091): the investigator reads it

```
read_section(summary) → "2 y#%rs%of e@%er%@%~e…"
propose(playwright)   → accepted (match 0.80)
search(API testing)   → "U~ed P@aywrig@t, Postman…"
propose(api_testing)  → accepted
— on syn_043:
propose(python)       → rejected (quote lacks python, 0.50)
```

Across 42 planted cases (qwen3:8b) and 17 (Sonnet 5.5): **0 false recoveries**. On the benchmark the investigator rescued no CV — correctly: of the knocked-out CVs it selects, 36 of 37 are graded 0–1.

6.3 · WHAT HR SEES

An explanation card that adds up — JD-010, CV-033

Criterion	w	score	contribution	evidence (quoted from the CV)
tools	0.20	92	0.184	must-haves 3/3 · "Xây dựng framework Selenium + Java + TestNG..."
automation depth	0.20	75	0.150	L3 from the candidate's own verbs
experience	0.10	100	0.100	8.8 years in automation roles (10.8 total)
test types	0.10	100	0.100	"~200 RestAssured tests for loan and account APIs..."
CI/CD and infra	0.10	90	0.090	GitLab CI, Docker; Jenkins only listed (0.7)
ISTQB · language · domain	0.10 ×3	100	0.300	"ISTQB Advanced" · Java · English & banking
total			**0.924**	

For a knocked-out candidate (CV-040): "Missing must-have: Selenium". **What if** Selenium were used? It would pass the knock-out and score 0.855 — HR sees exactly how close a rejected candidate was.

7 · HYPOTHESES · 8 · ERROR ANALYSIS

Scored against what we measured

Hypothesis (midterm)	Measured	Verdict
H1 agent beats the fixed pipeline	0.851 vs 0.839 (p=0.064)	**Not supported**
H2 selective arbiter cuts token cost	P6 ≈ 2.1× B4; vs LLM-every −47% / −67%	**Partly**
H3 verification keeps explanations grounded	100% grounded; claims 2.8% → 0.9%	**Met (different reason)**
H4 company profiles align with HR	τ 0.554 vs 0.542	**Not supported***
H6 asking HR helps on vague JDs	τ +0.056, nDCG +0.129 (p<0.001)	**Supported**

Error analysis (test)	ours (6)	coding (4)
P6 full agent	6 miss · 1 false alarm	4 miss · 1 false alarm
Rubric ranker	10 miss · 1	4 miss · 0

Two failure modes: vague JDs (asking HR recovers 57%) and CVs that *describe* tools without naming them (3 of 6 misses on ours). *H4 labels were written without company context.

9 · LIMITATIONS & ETHICS

Stated plainly

Human labels (ours)
120 pairs, κ 0.965; vs silver 0.66–0.70 → silver secondary. v1.1 keeps the order (τ 1.00); numbers stay synthetic.

Real-data pilot
Djinni 424 pairs (κ 0.696), an aggregate not an adjudication; one set saw the model grade; only 8 JDs.

Small test split
10 JDs → CIs ±0.05–0.08; differences under ~0.03 not significant.

Circularity firewall
`rank_shortlist`, P7 0.953 and S1 0.942 are not effectiveness claims; off until a frozen human-gold run.

The served LLM
Claude via the Claude Code CLI as an API stand-in; calls/tokens/behaviour carry over, latency & cost do not.

Ethics
Protected attributes deleted before indexing (twins: 0 change). The system recommends; HR decides, logged.

10 · CONCLUSION · 11 · CONTRIBUTIONS

Agentic where rules cannot reach

Ranking
Best version (P6 0.851); order holds when people grade (v1.1). Real CVs: BM25 0.422 → rules 0.636 → P6 0.643.

Agentic behaviour
P7 passes 40/42 (HR tasks) and 81/83 (complex goals) — returns to HR when a goal cannot be met.

Control
Where only same-family labels could grade an LLM ordering, the result is not claimed and the feature stays off.

It keeps the score in deterministic rules over well-processed data and checks every LLM output in code. Its evaluation found faults unit tests never would. Next: the locked Djinni holdout · E15 with repeated runs & a real HR · consented real CVs · a larger test split.

Contributions — Sơn: problem & criteria · Vũ: literature (31 papers) · Đán: agentic-HR, gaps G1–G5, coding set, 8-axis rules, both-order arbiter, explanation cards · Phúc: blind labels with Hoa · Hoa: datasets, metrics, validation · Thịnh: architecture, agent, pipeline, benchmark, experiments.

THANK YOU · Q&A

An agentic CV ranker that earns trust by being agentic where rules cannot reach